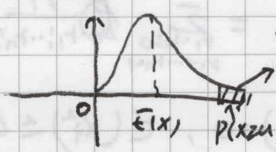
Recap on probability.

Concentration inequalities.

$\left|\frac{1}{n}\sum_{i=1}^{n} X_i - E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]\right|$ is small gareented with high probability.

Markov inequality.

X is non-negtive R.V., for $\forall a > 0$

$\mathbb{P}(X \geq a) \leq \frac{E[X]}{a}$ — 

the probability of X is too large $P(X \geq a)$ is bounded.

If X is continous R.V. with $f_X(x)$ as pdf

$$E(X) = \int_0^{+\infty} x f_X(x)\,dx = \int_0^{a} x f_X(x)\,dx + \int_a^{\infty} x f_X(x)\,dx$$

$$\geq \int_a^{\infty} x f_X(x)\,dx \geq a\int_a^{\infty} f_X(x)\,dx = aP(X \geq a).$$

Remarks:

+: almost always works (X non-negative)

−: Bond is week (not precise)

Power trick → sharper bound.

$P(X^r \geq a^r) \leq \frac{E(X^r)}{a^r}$ (assume r^{th} moment $E(X^r)$ exist).

Chebyshev's in equality.

X be R.V, for $\forall c > 0$

$\mathbb{P}(|X - E(X)| \geq c) \leq \frac{Var(X)}{c^2}$ — power trick of Markov

$$\mathbb{P}(|X - E(X)| \geq c) = P(|X - E(X)|^2 \geq c^2) \leq \frac{E((X-E(X))^2)}{c^2} = \frac{Var(X)}{c^2}.$$

Master tail bond.

Let $X = X_1 + X_2 + \cdots + X_n$, where $X_1, \ldots, X_n$ are i.i.d and

- $E(X_i) = 0$ • $E[X_i^2] \leq \sigma^2$ • $|E[X_i^s]| \leq \sigma^2 s!$ for $s = 3, 4, \ldots, \lfloor \frac{a^2}{4n\sigma^2} \rfloor$

and $a \leq \sqrt{2}n\sigma^2$.

$\Rightarrow \text{prob}(|X| \geq a) \leq 3e^{-\frac{a^2}{12n\sigma^2}}$

prof scheteh:

① chuse r an even integer s.t. $0<r\leq s$

$$P(|X|\geq a) = P(X^r \geq a) \leq \frac{E(X^r)}{a^r}$$

② $E(X^r) = E\left((X_1+X_2+\cdots+X_n)^r\right) = E\left(\sum_{\substack{r_1,\ldots,r_n\geq 0 \\ r_1+\cdots+r_n=r}} \frac{r!}{r_1!r_2!\cdots r_n!} X_1^{r_1}\cdots X_n^{r_n}\right)$

$= \sum_{\substack{r_1\cdots r_n\geq 0 \\ r_1+\cdots+r_n=r}} \frac{r!}{r_1!\cdots r_n!} E\left(X_1^{r_1}\cdots X_n^{r_n}\right)$ → use independence

③ $E(X_i)=0$, $E(X_i^2)\leq \sigma^2 \leq 2\sigma^2$, $E(X_i^{r_i}) \leq r_i!\,\sigma^2$

$\Rightarrow \sum_{\substack{r_1\cdots r_n\geq 0 \\ r_1+\cdots+r_n=r}} = \sum_{\left(\substack{r_1\cdots r_n\geq 0 \\ r_1+\cdots+r_n=r \\ \text{if } r_i\neq 0,\ r_i\geq 2}\right)} \Rightarrow t = \#$ of non zero r_i

$\Rightarrow 1\leq t\leq r/2$ (r_i are aleast 2)

④ Eenumarate $\sum_{(\ )}$ to enumarate t $\begin{cases} \text{fixed } t,\ C_n^t \text{ way to chuse } r. \\ \text{after } \{r_i\} \text{ fix},\ C_{r-2t+t-1}^{t-1} \text{ way.} \end{cases}$

* $\{r_i\}$ fixed, ~~first give~~ ev like puting ball into bins, first evey r_i has two, the rest $(r-2t)$ will be put arbitarily. use insertion methods to count total ways.

⑤ $E(X^r) \leq r!\sum_{t=1}^{r/2}\binom{n}{t}\binom{r-t-1}{t-1}\sigma^{2t}$, use $\begin{cases} \binom{n}{t}\leq \frac{n^t}{t!} \quad \left(\frac{n!}{t!(n-t)!}\leq\frac{n\cdot n\cdots n}{t!}\right) \\ \binom{r-t-1}{t-1}\leq 2^{r-t-1} \quad \left(2^n\geq\binom{n}{i},\ i=0,\ldots n \text{ binary expansion}\right) \end{cases}$

$\Rightarrow E(X^r)\leq r!\sum_{t=1}^{r/2} h(t)$. using $\frac{h(t)}{h(t-1)}\geq 2$

⑥ note $r\leq s\leq\left\lfloor\frac{a^2}{4n\sigma^2}\right\rfloor$ $\Rightarrow r\leq s\leq\frac{nr^2}{2}$

$0<a<\sqrt{2}\,n\sigma^2$

$\Rightarrow$ ~~$E(X^r)$~~ $P(|X|\geq a) \leq \left(\frac{2rn\sigma^2}{a^2}\right)^{r/2} = g(r)$

⑥ use the fact that $g(r)\downarrow$ as $r-1\leq\frac{a^2}{4n\sigma^2}$.
chuse r be the largest even integer. s.t. $r\leq\frac{a^2}{6n\sigma^2}$.

$$P(|X|\geq a)\leq (1/3)^{r/2}\leq e^{-r/2}\leq e\,e^{-a^2/12n\sigma^2}\leq 3e^{-a^2/12n\sigma^2}$$

→ Full prof see Appendix.

Remarks: Master tail bound is a general probability tail bound for sum of iid R.V. that gives tighter concentration than CLT. Tail bound for various distribution can be derived

Example: Chernoff Bounds. case of iid Bernoulli.

$Y_1, \cdots Y_n$ iid Bernoulli(p) $\rightarrow \begin{cases} 1, & p \\ 0, & 1-p \end{cases}$

let $Y = Y_1 + \cdots + Y_n$, then for $\forall c \in [0,1]$

$$P\left[\left|\frac{Y}{n} - p\right| \geq \sqrt{2}\, c p(1-p)\right] \leq 3e^{-\frac{np(1-p)c^2}{6}}$$

Apply Master tail bound to $X_i = Y_i - p$. $E(X_i) = 0$

$E(X_i^2) = E((Y_i - p)^2) = E(Y_i^2) + p^2 - 2pE(Y_i) = E(Y_i^2) - E^2(Y_i) = Var(Y_i)$

$|E(X_i^s)| = |p(1-p)^s + (1-p)(-p)^s| \leq p(1-p) = Var(Y_i)$.

Chernoff Bonds is obtained directly from master tail bond with $a = n\sqrt{2}\, cp(1-p)$

concentration inequality using moment generating function.

Apply Markov ~~to~~ not to $|X|^r$, but to $e^{\lambda X}$.

$$P(X \geq t) = P(e^{\lambda X} \geq e^{\lambda t}) \leq \frac{E[e^{\lambda X}]}{e^{\lambda t}}, \ \forall \lambda > 0.$$

$\Rightarrow P(X \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t} E[e^{\lambda X}]$, noticed $E[e^{\lambda X}]$ is moment generating function (MGF)

most lower bound.

Tail bond for Gaussian R.V.

$$X \sim N(0, \sigma^2), \quad E(e^{\lambda X}) = e^{\frac{1}{2}\sigma^2\lambda^2} \Rightarrow P(X \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t} e^{\frac{1}{2}\sigma^2\lambda^2} = e^{-t^2/2\sigma^2}.$$

Hoeffding's Theorem.

let X_i be indep R.V. with $X_i \in [a_i, b_i]$ a.s, then $\forall t \geq 0$, $Y = \sum_i X_i$

$$P(|Y - E(Y)| \geq t) \leq 2e^{-\frac{2t^2}{\sum_i (b_i - a_i)^2}}$$

Hoeffding's lemma:

If X be a bounded R.V. s.t. $X \in [a, b]$ a.s., then

$$E(e^{\lambda(X - E(X))}) \leq e^{\frac{1}{2}\left(\frac{b-a}{2}\right)^2 \lambda^2}$$

proof of Hoeffding's theorem:

use ~~chebyshev~~ Markov tail bound for exponential

$P(\sum_i x_i - E(x_i) \geq t) \leq e^{-\lambda t} E(e^{\lambda(\sum x_i - E(x_i))})$

$= e^{-\lambda t} \prod_i E(e^{\lambda(x_i - E(x_i))})$

use Hoeffding's lemma, $E(e^{\lambda(x_i - E(x_i))}) \leq e^{\frac{1}{2}\frac{(b_i-a_i)^2}{4}\lambda^2}$

$\Rightarrow P(\sum_i x_i - E(x_i) \geq t) \leq e^{-\frac{2t^2}{\sum_i (b_i-a_i)^2}}$ ← inf of

Same thing can be show for $-x_i$. as $P(\sum_i x_i - E(x_i) \leq -t)$.

$\Rightarrow P(|\sum_i x_i - E(x_i)| \geq t) \leq 2e^{-\frac{2t^2}{\sum_i (b_i-a_i)^2}}$.

χ^2 tail bound.

$z_i \overset{iid}{\sim} N(0,1)$ $\quad X = \sum_i z_i^2 \sim \chi^2(n)$, $E(X) = n$, then.

$P(|X - EX| \geq t) \leq 2\exp\left[-c\min(t, \frac{t^2}{n})\right]$, $\forall t \geq 0$

proof: $P(X - E(X) \geq t) \leq e^{-\lambda t} e^{-n\lambda} E(e^{\lambda \sum_i z_i^2})$

$= e^{-\lambda t} e^{-n\lambda} \prod_i E(e^{\lambda z_i^2})$

$E(e^{\lambda z_i^2})$ is mgf of $\chi^2(1) = (1-2\lambda)^{-1/2}$, $\lambda \in (0, 1/2)$.

$\Rightarrow P(X - E(X) \geq t) \leq e^{-\lambda t} e^{-n\lambda} (1-2\lambda)^{-n/2}$.

$= e^{-\lambda t} (\underbrace{e^{-\lambda}(1-2\lambda)^{-1/2}}_{\frac{1}{e^{\lambda}\sqrt{1-2\lambda}}})^n \leq e^{-\lambda t + 2n\lambda^2}$, $\forall \lambda \in (0, 1/4)$

$\frac{1}{e^{\lambda}\sqrt{1-2\lambda}}$ increase as $\lambda \to 1/2$.

$e^{2\lambda^2} > \frac{1}{e^{\lambda}\sqrt{1-2\lambda}}$, $\lambda \in (0, 1/4)$

$\min_{\lambda \in (0, 1/4)} e^{-\lambda t + 2n\lambda^2} = \exp(-c\min(t, \frac{t^2}{n}))$

$\Rightarrow P(X - E(X) \geq t) \leq \exp(-c\min(t, \frac{t^2}{n}))$

$P(X - n \leq -t) = P(-(X-n) \geq t) \leq e^{-\lambda t} e^{\lambda n} E e^{-\lambda X} = e^{-\lambda t} e^{n\lambda}(1+2\lambda)^{-n/2}$.

$\leq e^{-\lambda t} e^{n\lambda^2}$ $\quad \forall \lambda > 0$

$P(X - n \leq -t) \leq e^{-\frac{t^2}{4n}}$

$\Rightarrow P(|X - E(X)| \geq t) \leq 2\exp(-c\min(t, \frac{t^2}{n}))$.